

Advanced Machine Learning Approaches for Humor Detection

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Brief Description of the Proposed Project:

Research Question: Can we develop a deep learning model tailored for humor classification that achieves better performance than existing models designed specifically for humor detection?

Motivation: Humor detection is a complex classification task that goes beyond surface-level text understanding. It requires recognizing linguistic nuances, such as incongruity, puns, sarcasm, and cultural references. While current state-of-the-art models show strong performance, there remains room for improvement through novel architectures. Our goal is to develop a model that better captures the nuanced linguistic patterns that make text humorous.

Current Knowledge: Recent advances in NLP and neural architectures—such as CNNs, attention mechanisms, and transformer-based models like BERT and RoBERTa—have significantly improved performance in text classification tasks. These models have also been applied to humor detection, with works like “Humor Detection: A Transformer Gets the Last Laugh” - Weller & Seppi (2019) and “Humor recognition using deep learning” - Chen & Soo (2018) demonstrating promising results on joke datasets. Reddit’s r/Jokes offers large labeled datasets, but most models remain general-purpose, leaving room for humor-specific improvements.

Approach: We will implement and systematically compare multiple approaches for binary humor classification:

- Implement other baselines from other papers for this specific task.
- Implement a new classifier (pBased on novel architecture)
- Compare baselines with other state-of-the-art classifiers and our implementation
- Conduct cross-dataset evaluation to assess generalization capabilities

Evaluation Strategy: We will use established humor detection datasets including:

Short Jokes dataset and 16000 One-Liners dataset

Evaluation metrics will include Accuracy, Precision, Recall, and F1-Score.

Success Criteria:

- Achieve competitive or superior performance compared to current state-of-the-art baselines on humor detection datasets
- Provide analysis demonstrating our model’s advantages and insights into humor-specific linguistic patterns

Assumptions and Requirements:

- **Computational:** GPU access through Tel-Aviv University Data Center Slurm cluster (provided)
- **Data:** Publicly available humor datasets (no additional labeling required)
- **Software:** PyTorch, HuggingFace Transformers, standard NLP libraries
- **Storage:** Dedicated storage path for models and datasets (provided)